Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Agent-Driven Workflows · AI Evaluation · SDLC Automation · Python · Azure · LangGraph · LangChain · Observability · 9yr Production

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Software engineer with 9+ years of production experience specializing in agent-driven workflows, AI evaluation frameworks, and SDLC automation - the exact capabilities Microsoft's AI Engineering Excellence team is building at scale. AgentProbe is the direct proof: I designed and shipped a production AI evaluation framework end-to-end - CI-integrated regression gate blocking quality drift before deployment, structured tracing across all agent steps, offline batch evaluation pipelines, and a 5-metric quality engine measuring correctness, groundedness, hallucination rate, latency, and token efficiency. This is agent-driven quality assurance applied to the software lifecycle. LeaseI demonstrates the agent-driven workflow pattern at production scale: an 8-agent LangGraph StateGraph automating a complex multi-step investigation workflow from end to end, with measurable outcomes (93.6% precision, 87.9% F1 across 1,240 test cases). I build with LangGraph and LangChain daily, use Claude Code as my core engineering workflow, and have hands-on Azure cloud experience from graduate research at George Mason University. At Entrata I owned production systems for 7 years - writing design documents, leading projects across multiple engineers, and measuring outcomes to drive continuous improvement. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agent-Driven Engineering & AI: Python (9yr), LangGraph, LangChain, MCP (Model Context Protocol), AI-Native Engineering, Agent-Driven Workflows, SDLC Automation, Agentic AI, Multi-Agent Orchestration, Tool Calling, HITL Guardrails, Prompt Engineering, Anthropic API, OpenAI API, Ollama, Claude Code, Generative AI, AI Coding Agents, Output Validation, Confidence Scoring

Evaluation, Observability & Testing: Evaluation Frameworks, AI Evaluation, LLM-as-Judge, Hallucination Detection, Groundedness, pass@k Harness, CI Regression Gate, Drift Detection, Structured Logging, Observability, Tracing, Monitoring, Alerting, Test Automation, Regression Testing, Test Harness Development, GitHub Actions CI/CD, Automated Testing, Feedback Loops

Platform & Infrastructure: Azure Cloud, AWS (S3, EC2, RDS, Glue, Lambda), FastAPI, REST APIs, Docker, Microservices, RabbitMQ, PostgreSQL, MySQL, Oracle, Redis, ChromaDB, Vector Databases, RAG Pipelines, React.js, TypeScript, Linux, Bash, Git, Design Documents, Technical Mentoring, Code Reviews, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Agent-Driven Workflow Delivery

- Led 0→1 delivery of a production agent-driven workflow (LangGraph, FastAPI, Python) as sole technical owner - translated ambiguous operational requirements into a working system, validated with the customer, and deployed to production in weeks; built evaluation pipeline and CI validation gate; delivered tracing and observability infrastructure for ongoing operation. Measured outcomes drove iteration decisions throughout.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on Azure cloud infrastructure; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems; delivered technical output across 3 research pipelines with cross-functional stakeholder reporting.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Production software & developer tooling** - owned mission-critical Python backend systems for 7 years at 99.5% uptime; built internal debugging tools reducing engineering triage time 40%; Python/Bash automation eliminating 80% of manual workload; RabbitMQ async processing with retry logic and idempotency guarantees; Docker microservices redesign (50% scalability improvement); 200+ SQL optimizations (30% performance gain); GitHub Actions CI/CD with automated testing; AWS S3/EC2/RDS/Glue.
- **SDLC practices & engineering excellence** - wrote design documents and architecture decisions for 20+ production modules; led projects spanning multiple engineers; mentored 5+ engineers on system design and production practices; 300+ peer code reviews; built meaningful monitoring, alerting, and runbooks; incident response with RCA; kept bug count below 10 for 6+ consecutive months across 120+ Agile sprints - measurable engineering quality improvement across the team.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL); data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: AI Evaluation Framework - Agent-Driven SDLC Quality Gate

AI Evaluation · CI Gate · Tracing · SDLC Automation · Open Source

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation framework applying agent-driven quality assurance to the software development lifecycle - the pattern CEAI's AI Engineering Excellence team scales across Microsoft. **CI-integrated regression gate:** GitHub Actions integration blocks quality drift before deployment, making evaluation a first-class SDLC engineering practice. **Evaluation engine:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k multi-trial harness surfacing variance in non-deterministic agent outputs. **Observability:** structured tracing across all agent steps; offline batch evaluation pipelines; multi-provider LLM routing (Anthropic, OpenAI, Ollama) with cost and performance tradeoffs. Open-source, multi-team deployable. Zero regressions across 1,240+ test cases.

Leasel: Agent-Driven Workflow - Production Multi-Agent SDLC Pattern

Multi-Agent · LangGraph · SDLC Automation · Measurable Outcomes · Production

Python · LangGraph (8-agent) · FastAPI · RAG · SQLite · MCP · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production multi-agent system demonstrating the agent-driven workflow pattern CEAI is building for the software lifecycle. **Workflow automation:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) automates a complex multi-step investigation workflow end-to-end with conditional routing and dynamic task delegation. **Measured outcomes:** 93.6% precision, 87.9% F1 across 1,240 test cases - the measurement-driven approach that turns experiments into repeatable practices. **Human oversight:** deterministic HITL escalation enforces safe execution on regulated decision paths - appropriate human oversight baked into agent architecture.

AI on FHIR: Full-Stack Production System - Azure-Compatible Cloud Architecture

Cloud · AWS · REST APIs · HIPAA · Production

Python · Flask · React.js · TypeScript · Docker · AWS · Azure-compatible · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack production AI system on cloud infrastructure - LLM-powered tool invocation translating natural-language queries into structured API calls against regulated healthcare data; HIPAA-compliant RBAC with full data isolation and audit trail; REST API backend with Docker containerization and cloud-native deployment; ~60% reduction in manual retrieval time. Demonstrates secure AI system design, cloud infrastructure, and production observability in a regulated environment.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87· Azure Cloud Research

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017